Hopf-Charge Bounds for Nodal States in Conditional Berger–Hopf Light Transfer

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Research article draft for *Galaxies*: Strong Gravitational Lens Modeling

Abstract

A scalar preparation can leave a fiducial light ray undeflected at first order while deforming its neighboring beam. We determine the Hopf content required by this nodal construction in the degree-two scalar sector of the three-sphere. For a nonzero normalized quadratic harmonic whose value and full spatial gradient vanish on a great circle, the squared-norm fraction in absolute Hopf charge two is exactly $(1+\kappa^2)/2 \geq 1/2$, where κ is the absolute Kähler cosine of the ray plane. A Hopf-fibre ray requires a purely charged state. The result fixes its Berger Dirichlet Rayleigh quotient and gives a rigidity criterion for persistence of the node under an explicitly specified scalar spectral action. We apply the normalized nodal family to a separately declared round-background scalar–tensor realization used in the Berger–Hopf program. Its full six-coordinate state determines incoming, outgoing and observer–source Jacobi responses, fixed-physical-source timing shifts, and an evaluated antisymmetric ordered-transfer kernel for independent angular histories. The kernel is absent in a common-profile preparation. General lensing theory nevertheless permits a common affine source transformation that symmetrizes the strong-lens maps in the tidal approximation. The advance is a sharp geometric restriction on the preparation and a reproducible conditional response family; it supplies neither a microscopic action-to-cosmology identification nor a unique observable amplitude.

Keywords: strong gravitational lensing; Hopf fibration; Berger sphere; scalar harmonics; Jacobi map; time delay

1 Physical question and scope

The Berger–Hopf Standard Model (BHSM) research program organizes physical responses around envelope geometry, boundary matching and a constrained action. In a lensing application, an effective gravitational potential may represent the same contribution as an envelope description. A useful result therefore need not depart from general relativity, but it must determine something beyond assigning an arbitrary angular profile and amplitude.

The starting angular calculation used normalized degree-two harmonics of a round closed background and a specified six-coordinate scalar–fluid state [1, 2]. One preparation has a central spatial ray on which the scalar value and its first derivatives vanish, while a nonzero transverse Hessian changes the beam. The resulting fixed-source singular-isothermal-sphere (SIS) delay is linked to incoming and total-path shear. That is a conditional effective-model calculation. A quadrupolar $\cos 2\varphi$ pattern, by itself, does not establish a Hopf-fibre effect or a physical state selection.

We ask a more restrictive question: *what Hopf representation content must that nodal preparation have?* Theorem 1 answers it without choosing a numerical amplitude. Its sharp half-power bound holds for every orthogonal Hopf structure; the precise fraction depends on the relative orientation

of the ray plane. A second theorem states exactly when the same node can persist for a free scalar on a fixed Berger space. These are spatial geometry and explicitly specified scalar-action results. The cosmological optical application remains on the separate round branch.

The scalar Berger spectrum is known [3]. The harmonic and full-sky shear machinery is established [4, 5]. Distance-matrix lens equations, arrival times and equivalent-lens freedom are also established [6–8]. Our addition is the sharp constrained charge fraction, its nodal-persistence consequence, and the evaluated state response for the declared optical realization. The general ordered-tide rotation mechanism is already derived in the literature, including a closely related recent Jacobi-map treatment [9]; we use it as a checkable response channel.

No unique microscopic BHSM reduction to the chosen scalar–tensor action is assumed. The recovered boundary/Hessian machinery identifies concrete common-domain projection requirements, but does not evaluate that bridge. The microscopic fixed-external-label condition also does not impose alignment of cosmological angular states. These distinctions motivate the state family used here. The source archive contains the finite recovery map, hashes and reconciliation, rather than treating an unlocated object as a proof that it was never derived.

2 Normalized spatial state and Hopf structure

Let S^3 be the unit sphere embedded in $\mathbb{R}^4$, with coordinates X. Write the physical round spatial metric as $R^2 h_1$, $K = R^{-2}$, and consider a real symmetric tracefree matrix A. Define

$$F_A(X) = X^T A X, \qquad Q_A = \mathcal{N} F_A, \qquad \mathcal{N} = \sqrt{6/(\pi^2 R^3)}. \tag{1}$$

Restriction of a homogeneous harmonic quadratic gives $-\Delta_{R^2 h_1} Q_A = 8 K Q_A$. The fourth moment of normalized round measure is

$$\int X_i X_j X_k X_l \, d\mu = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/24. \tag{2}$$

Consequently $\int F_A F_B d\mu = \mathrm{tr}(AB)/12$ and $\int Q_A Q_B dV_{R^2 h_1} = \mathrm{tr}(AB)$. Frobenius-orthonormal STF matrices thus give nine orthonormal scalar harmonics. Observer-centered sky decomposition gives one monopole, three dipoles and five quadrupoles; the degree $n = 2$ on S^3 must be distinguished from sky angular degree.

An orthogonal complex structure J satisfies $J^T = -J$, $J^2 = -I$. Its Hopf flow is $X \mapsto e^{\theta J} X$, generated by $\xi = JX$. The orthogonal involution $A \mapsto J^T A J$ gives

$$A_0 = \tfrac{1}{2}(A + J^T A J), \qquad A_2 = \tfrac{1}{2}(A - J^T A J). \tag{3}$$

The neutral subspace has real dimension three; it obeys $[A_0, J] = 0$ and $\xi F_{A_0} = 0$. The charged subspace has dimension six, obeys $\{A_2, J\} = 0$ and $\xi^2 F_{A_2} = -4 F_{A_2}$. The latter real space contains the charges $q = \pm 2$ together. The power fraction

$$f_2(A; J) = \|A_2\|_F^2 / \|A\|_F^2 \tag{4}$$

is therefore a normalized scalar L^2 fraction, not an observed mass fraction or a probability distribution over envelopes. At the round point a preferred J is not dynamically selected by isotropy.

3 The stationary-nodal charge bound

Let P be a real two-plane and $W = P^\perp$. Its great circle is $P \cap S^3$. We require both $F_A = 0$ and the *full* spherical gradient $\nabla F_A = 0$ on that circle. Requiring only a derivative tangent to the circle would impose no such restriction.

Theorem 1 (Sharp charge fraction)**.** *For nonzero symmetric tracefree A stationary-nodal on $P \cap S^3$, let e_1, e_2 be an orthonormal basis of W and set $\kappa = |e_1^T J e_2|$. Then*

$$\boxed{f_2(A; J) = \frac{1+\kappa^2}{2} \in [1/2, 1].} \tag{5}$$

The absolute Kähler cosine of P equals that of W in real dimension four. The result is independent of the transverse amplitude and orientation of A. The upper endpoint holds exactly when P is J-invariant, so that its nodal circle is a Hopf fibre. The lower endpoint is attained by a totally real plane.

Proof. On the unit sphere, $\nabla F_A = 2(AX - F_A X)$. Simultaneous vanishing of value and gradient gives $AX = 0$. An open arc spans P, so $A|_P = 0$. Symmetry and trace freedom then give, in the splitting $W \oplus P$,

$$A = C \oplus 0, \qquad C = \begin{pmatrix} p & q \\ q & -p \end{pmatrix} \neq 0. \tag{6}$$

The compression of the skew matrix J to W is $J_{WW} = j \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, where $|j| = \kappa$. Direct multiplication gives

$$\operatorname{tr}(AJ^T AJ) = \operatorname{tr}(C J_{WW}^T C J_{WW}) = -\kappa^2 \operatorname{tr}(C^2). \tag{7}$$

Since the involution preserves the Frobenius norm, $\|A_2\|^2 = (\|A\|^2 - \langle A, J^T A J\rangle)/2$, proving Eq. (5). Orthogonality of J gives $\kappa \leq 1$. Equality implies $J(W) = W$, equivalently $J(P) = P$. Conversely such invariance gives $\kappa = 1$. Totally real P has $\kappa = 0$. Both endpoints admit nonzero matrices (6). □

The theorem is stronger than a chosen "Hopf-neutral" restriction: no Hopf orientation makes this nonzero nodal family neutral. Every instantaneous nonzero combination of the two transverse tensors in Eq. (6) has the same fraction for fixed P, J. Independent temporal histories therefore do not invalidate the spatial bound. This applies to the fixed nodal subspace; finely adjusted cancellations along a spacetime trajectory with unrelated time-varying spatial tensors are not covered by the hypothesis.

3.1 Exact spectral consequence and scale convention

Use the Berger family

$$h_{a,R} = R^2(h_H + a^{-2}\eta^2), \qquad -\Delta_{a,R} = R^{-2}[-\Delta_1 - (a^2-1)\xi^2], \quad a > 0. \tag{8}$$

Here a is the inverse relative fibre scale and R the horizontal radius, distinct from the cosmological scale factor below. The known scalar spectrum is $R^{-2}[k(k+2) + (a^2-1)q^2]$ [3]. Thus $n = 2$ splits into $8/R^2$ and $(4+4a^2)/R^2$. The volume differs from round volume by the constant a^{-1}, so normalized measure and power fractions are unchanged.

For fixed A, R the exact Dirichlet Rayleigh quotient is

$$\boxed{\mathcal{R}_a[A] = \frac{8 + 4(a^2-1)f_2}{R^2}, \qquad \left.\frac{\partial \mathcal{R}_a}{\partial(a^2)}\right|_{A,R} = \frac{2(1+\kappa^2)}{R^2}.} \tag{9}$$

This derivative lies between $2/R^2$ and $4/R^2$. A mixed-sector function is generally not an eigenfunction when $a \neq 1$; the quotient must not be relabeled an eigenvalue. For $a > 1$, the quotient lies between $(6+2a^2)/R^2$ and $(4+4a^2)/R^2$; their ordering reverses for $a < 1$. All reunite at $8/R^2$ when $a = 1$.

The held scale matters. Fixed volume instead requires $R(a) = R_0 a^{1/3}$; at $a = 1$ the derivative with respect to a^2 becomes $(4f_2 - 8/3)/R_0^2$, which can be negative. Equation (9) is a fixed-horizontal-radius statement, not a scale-independent lower bound on the full BHSM envelope energy. Other terms in a physical action may modify its energy and dynamics.

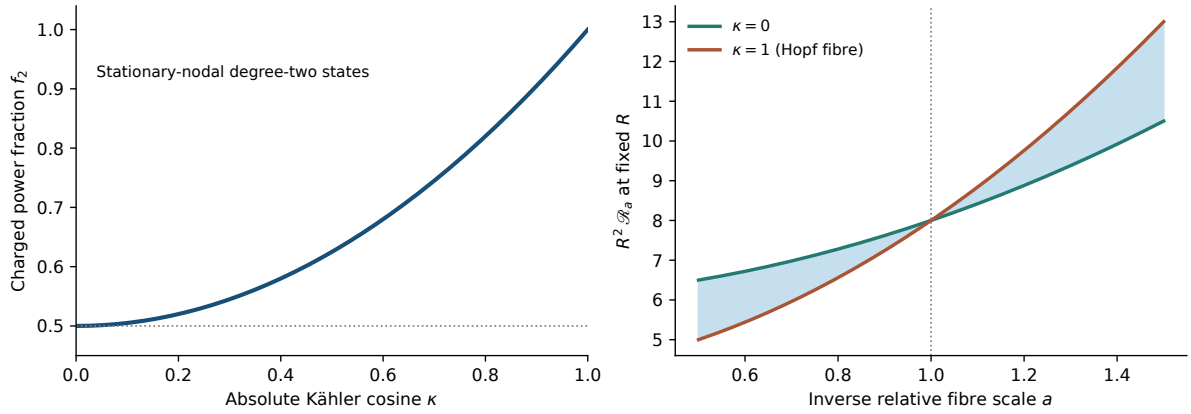


Figure 1: Exact geometric restriction, with no fitted physical amplitude. Left: mandatory charged fraction against κ. Right: fixed-horizontal- radius Dirichlet Rayleigh quotient for the stationary-nodal family. The shaded region spans all Hopf orientations and is not a confidence band. Neither panel computes squashed-spacetime light propagation.

3.2 A specified action and persistence of the node

An auxiliary test action makes the dynamical limitation explicit:

$$S_{\text{test}} = \tfrac{1}{2} \int dt \left[\|\dot{F}\|^2_{L^2(d\mu)} - \langle F, -\Delta_{a,R} F \rangle_{d\mu} \right]. \tag{10}$$

It describes a free scalar on a fixed product time–Berger space, in units with wave speed one. Variation gives $\ddot{F} - \Delta_{a,R} F = 0$. The degree-two subspace is invariant under this exact operator, so its finite reduction is justified without an arbitrary projected stiffness. This action is an explicit auxiliary realization, not the cosmological R1 action or a derived microscopic envelope action.

Theorem 2 (Nodal persistence)**.** *For $a \neq 1$, a nonzero degree-two solution of Eq. (10) can remain stationary-nodal on one fixed great circle throughout a time interval only if that circle is a Hopf fibre. If it is a Hopf fibre, its nodal subspace is invariant.*

Proof. Write $\mathcal{V}(P) = \{A = A^T, \operatorname{tr} A = 0, A|_P = 0\}$. If $A(t)$ stays in this linear space then $\ddot{A}(t)$ does too. The equation and $a^2 - 1 \neq 0$ imply $A_2(t) \in \mathcal{V}(P)$, hence $A_0(t) \in \mathcal{V}(P)$. Theorem 1 excludes any nonzero neutral element of $\mathcal{V}(P)$. Thus every nonzero $A(t)$ is entirely charged, and Eq. (5) forces $\kappa = 1$. Conversely $\kappa = 1$ makes every element entirely charged, on which the Laplacian is a scalar, preserving the subspace. □

The round case is an essential exception. For $a = 1$ both sectors evolve with the same operator, and every nodal plane is allowed. The zero solution and isolated nodal instants do not contradict the theorem. Operators that mix sectors, change J or the ray plane, or have additional field constraints require their own analysis. Thus persisting with the old nodal ansatz after squashing is a physical restriction, not an automatic continuation of round cosmology.

4 Conditional round-background action-to-light map

We next use the previously specified R1 action as a concrete optical realization on the round closed background. In code units $M_{\text{Pl}} = H_0 = c = 1$,

$$S_{\text{R1}} = \int d^4x \sqrt{-g} \left[\frac{\mathcal{R}}{2} + X - V_0 + q_T T - \frac{2X}{\lambda} \Box T \right] + S_m + S_r, \qquad X = -\tfrac{1}{2} (\nabla T)^2. \tag{11}$$

This is an established kinetic-gravity-braiding action class [10, 11], with minimally coupled irrotational dust and radiation. Its parameters are fixed: $(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}, \lambda, q_T) = (0.31, 9 \times 10^{-5}, -0.018, 1, 3)$ and $V_0 = 2.3986073449708067$. They are not fitted to lens data. Appendix A supplies the background, full finite action, constraints, normalization and canonical equations.

In uniform-field gauge, the auxiliary lapse, shift and fluid velocities are algebraically solved using their actual action block $A_{\rm aux}$. The reduced kinetic block is $M = M_0 - B^T A_{\rm aux}^{-1} B$. The physical degree-two mode has no homogeneous or pure-gauge dipole direction left to invert. The six-dimensional canonical state is (x, p) with $x = (\zeta, D_m, D_r)$ and $p = a_{\cos}^3(M\dot{x} + Lx)$, where $a_{\cos} = e^N$. No pseudoinverse of a gauge zero mode is used. Nonflat scalar constraints must be retained; replacing this reduction by a flat sound-speed equation would not follow from the closed Horndeski action [12].

At $z_i = 2.1$ the specified physical coordinates are

$$y = (\zeta, \Pi_{2,\rm ref}, \delta_m^{\rm ADM}, \delta_r^{\rm ADM}, v_m^{\rm ADM}, v_r^{\rm ADM})_i^T. \tag{12}$$

The reference Π is not the full coupled canonical momentum. An invertible six-row map $\mathcal{B}_i$ converts the physical anchor to the canonical state. The action fixes $U' = F(N)U$, $U(N_i) = I$, and the Newtonian reconstruction rows fix

$$w(N) = (\phi + \psi)(N)\, U(N, N_i)\mathcal{B}_i^{-1}, \quad ds^2 = a_{\cos}^2[-(1+2\Phi)d\eta^2 + (1-2\Psi)\gamma_{ij}dx^i dx^j]. \tag{13}$$

Thus $w \cdot y$ is the Weyl-potential coefficient of one normalized harmonic. The same temporal transfer applies to each of nine independent spatial harmonics on the round branch. This gives 54 initial coordinates, not an automatically aligned single profile. No action principle here selects these initial data.

4.1 Nodal tensors and three propagation segments

For a fiducial observer–radial ray take $P = \mathrm{span}(e_0, e_3)$ and parallel transverse screen (e_1, e_2). Let $Z = \mathrm{diag}(1, -1)$, $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and choose unit tensors $A_+ = (Z/\sqrt{2}) \oplus 0$ and $A_\times = (X/\sqrt{2}) \oplus 0$ on $W \oplus P$. Their screen Hessians are

$$(D_\perp D_\perp Q_+)_{\rm TF} = \sqrt{2}\mathcal{N}KZ, \qquad (D_\perp D_\perp Q_\times)_{\rm TF} = \sqrt{2}\mathcal{N}KX. \tag{14}$$

Both have zero value, full first derivative and screen Hessian trace along the central ray. First-order radial/frequency, endpoint and local frame terms therefore vanish for this preparation, while the beam deforms. These properties are spatial statements on the round branch.

For independent six-coordinate inputs u, v in these two tensors, define $S_K(r) = \sin(\sqrt{K}r)/\sqrt{K}$. For vertices $a < b$, the background physical angular-diameter distance is $d_{ab} = a_{\cos,b} S_K(\chi_b - \chi_a)$. The action-derived shear is

$$D_{ab} = d_{ab}(I - \Gamma_{ab}) + O(y^2),$$
$$\Gamma_{ab} = \sqrt{2}[(g_{ab} \cdot u)Z + (g_{ab} \cdot v)X], \tag{15}$$
$$g_{ab} = \frac{\mathcal{N}K}{S_K(\chi_b - \chi_a)} \int_{\chi_a}^{\chi_b} S_K(\chi_b - t) S_K(t - \chi_a) w(t)\, dt. \tag{16}$$

Lengths use c/H_0; the volume-normalized Q and field coefficients use those same units. No extra observer–vertex scale factor is added. All three maps ol, os, ls use the same state, but each has its own vertex conditions. In particular $D_{ls} \neq D_{os} - D_{ol}$.

This completes the conditional optical chain: the supplied effective action fixes a constrained, normalized propagator; its metric rows and the exact spatial Hessian fix incoming/local/outgoing transfer. The geometric theorem restricts the allowed nodal shapes. The missing microscopic identification is an earlier arrow, which this calculation does not replace with a free coefficient or a second copy of gravity.

5 Independent histories and ordered transfer

With $c_* = \sqrt{2}\mathcal{N}K$, the conformal screen equation is

$$D'' + KD = -\epsilon\mathcal{T}_1 D + O(\epsilon^2), \qquad \mathcal{T}_1 = c_*[(w\cdot u)Z + (w\cdot v)X], \quad D(a) = 0,\ D'(a) = I. \tag{17}$$

Here ϵ scales both inputs. In a consistently parallel orthonormal screen, every optical tidal coefficient is symmetric. Expanding $D = S_K I + \epsilon D_1 + \epsilon^2 D_2 + \cdots$, an arbitrary second-order tidal term multiplies $S_K I$ and contributes only a symmetric part to D_2. The ordered product of first-order tides gives

$$\operatorname{Asym}\frac{D_{ab}}{S_K(b-a)} = \epsilon^2(u^T \mathsf{B}_{ab} v)J_s + O(\epsilon^3), \quad J_s = ZX, \tag{18}$$

$$(\mathsf{B}_{ab})_{ij} = \frac{c_*^2}{S_K(b-a)}\int_a^b dt \int_a^t ds\, S_K(b-t)S_K(t-s)S_K(s-a) \times [w_i(t)w_j(s) - w_j(t)w_i(s)]. \tag{19}$$

Thus $\mathsf{B}^T = -\mathsf{B}$. If $v = ru$, all tides commute and the rotation vanishes exactly in the finite first-order-tide ODE. The converse need not hold: independent histories can cancel in a particular integral. Numerical singular values are basis-dependent and do not define a physical energy norm.

Equation (19) is the known ordered-tide mechanism, now evaluated on the normalized R1 state [9]. It describes map antisymmetry, not optical twist: $D^T D' - D'^T D = 0$, so $D'D^{-1}$ is symmetric where defined. Fixed-redshift endpoint shifts add symmetric terms at this order. A separately rotated source screen changes the matrix representation and must not be counted as a new signal. The independence from unknown second-order tidal terms applies to an isotropic *smooth* baseline. It does not give the complete second-order response after an order-one anisotropic local lens.

For numerical evaluation, the first-order positive-response row G and unnormalized B obey

$$G'' + KG = c_* w S_K(t-a), \qquad B'' + KB = c_*(wG^T - Gw^T), \tag{20}$$

with zero initial data, and $\mathsf{B} = B(b)/S_K(b-a)$. Independent ordered quadrature and direct matrix-ODE integration test this reduction. An even-amplitude projection isolates the quadratic coefficient. For the finite tracefree-tide ODE only even orders have antisymmetric parts. A Volterra majorant bounds higher even terms by

$$\frac{L}{S_K(L)}\sum_{m\ge 2}\frac{\nu^{2m}}{(4m+1)!}, \qquad \nu = L^2 \sup\|\mathcal{T}_1\|, \quad L = b-a, \tag{21}$$

using $0 \le S_K(r) \le r$ before the first conjugate point. It controls the declared optical equation; it is not a bound on omitted nonlinear field equations. The supplied numerical maximum is sampled rather than interval-certified.

6 Strong-lensing timing and representation freedom

For a fixed physical local potential $\widehat{\psi}$, source displacement X_s and measured redshifts, standard distance-matrix theory gives

$$\beta = D_{os}^{-1}X_s, \qquad \mathsf{T} = (1+z_l)D_{ol}^T D_{ls}^{-1} D_{os},$$
$$ct(\theta) = \tfrac{1}{2}(\theta-\beta)^T\mathsf{T}(\theta-\beta) - (1+z_l)\widehat{\psi}(D_{ol}\theta) + f(\beta). \tag{22}$$

The matrix T is symmetric by optical reciprocity [7]. For $D_{ab} = d_{ab}(I + E_{ab})$, $\delta\mathsf{T}/\mathsf{T}_0 = E_{ol} + E_{os} - E_{ls}$. Stationarity of an image removes its positional variation from the first derivative of arrival time. With $q_i = \theta_i - \beta$,

$$\frac{c\,\delta t_i - \delta f}{T_0} = \tfrac{1}{2}q_i^T(E_{os} - E_{ol} - E_{ls})q_i + q_i^T(E_{os} - E_{ol})\beta. \tag{23}$$

This includes the changing source angle and impact in the fixed physical lens. Holding angular source position fixed would give a different law.

For an SIS $\widehat{\psi}(x) = b|x|$, physical source offset se, $e = (\cos\varphi, \sin\varphi)$, and noncritical two-image branches, the quadratic term cancels between images. Therefore

$$\begin{aligned}\Delta t_0 &= \frac{2(1+z_l)bs}{c}\frac{d_{ol}}{d_{os}},\\ \frac{\delta\Delta t}{\Delta t_0} &= \sqrt{2}[(g_{os}-g_{ol})\cdot u\cos 2\varphi + (g_{os}-g_{ol})\cdot v\sin 2\varphi].\end{aligned} \tag{24}$$

The ls shear cancels from this particular timing derivative, while remaining in image positions. The earlier common + seed is recovered by setting $v = 0$. The angular pattern is inherited scalar shear geometry; the geometric result here concerns its mandatory Hopf content.

There is also a leading rotation in a chosen angular source chart. For a prescribed fixed symmetric reduced local-lens Hessian L, put $F = \Gamma_{ol} + \Gamma_{os} - \Gamma_{ls}$. The angular-source Jacobian $A_{\mathrm{ang}} = I - D_{os}^{-1} D_{ls}\widehat{\psi}'' D_{ol}$ has

$$\delta A_{\mathrm{ang}} = -(\Gamma_{os} - \Gamma_{ls})L + L\Gamma_{ol}. \tag{25}$$

Resampling a general lens at a shifted impact also adds a symmetric $-\delta L$, leaving the displayed antisymmetric part unchanged. The Jacobian can be nonsymmetric at first order because the main lens is already anisotropic at order zero. This is distinct from the quadratic smooth rotation in Eq. (18).

Neither raw antisymmetry supplies a source-independent identification. For common transfer matrices across all images, the physical-source Jacobian at image i is $A_i = D_{os} - D_{ls}U_i D_{ol}$, with symmetric U_i. One common affine source transformation gives

$$\underbrace{D_{ol}^T D_{ls}^{-1}}_{\mathcal{S}} A_i = D_{ol}^T D_{ls}^{-1} D_{os} - D_{ol}^T U_i D_{ol}, \tag{26}$$

which is symmetric for every image. This is the equivalent-lens freedom [6, 7], and is consistent with $\mathsf{T}A_{\mathrm{ang}}$ being a Fermat Hessian. The effective potential represents the same surface contribution; adding another copy would double count it. Restricted source/lens calibration or genuinely ray-dependent transfers are required to identify additional physics. Mass-sheet and source-position freedoms remain [8, 13]; multiple source redshifts alone do not remove them.

7 Executed checks and limits

The geometric proof is exact. The package independently verifies the projector dimensions, trace identity, full-gradient nodal condition and Berger operator action symbolically, and tests both orientation classes of J numerically. It includes endpoint examples and spectral values on both sides of the round point. The auxiliary-action check compares its exact two-sector evolution with the claimed nodal persistence criterion.

The recovered nine-stage angular calculation was replayed from its original provider hashes. Its chosen matter coefficient $u = (0, 0, 10^{-3}, 0, 0, 0)$, $v = 0$, $\sigma_v = 200$ km/s and fixed $s = 0.8$ kpc give $\Delta t_0 = 7.753621449$ days and $\delta\Delta t = -0.353216090\cos 2\varphi$ s at $(z_l, z_s) = (0.3, 1)$. These are forward-model values, not measured amplitudes.

For independent nodal histories the new kernel retains all fifteen distinct coordinate-pair coefficients in its machine-readable output. Two illustrations use $u = 10^{-3}e_2$ and either $v = 10^{-3}e_3$ (matter/radiation density) or $v = 10^{-3}e_1$ (reference scalar momentum), with zero-based coordinate indices from Eq. (12). At $z_s = 1$ the smooth normalized rotation coefficients are, respectively,

$$\omega_{mr}^{(2)} = -9.98254\times 10^{-19}, \qquad \omega_{m\Pi}^{(2)} = 6.70644\times 10^{-14}. \tag{27}$$

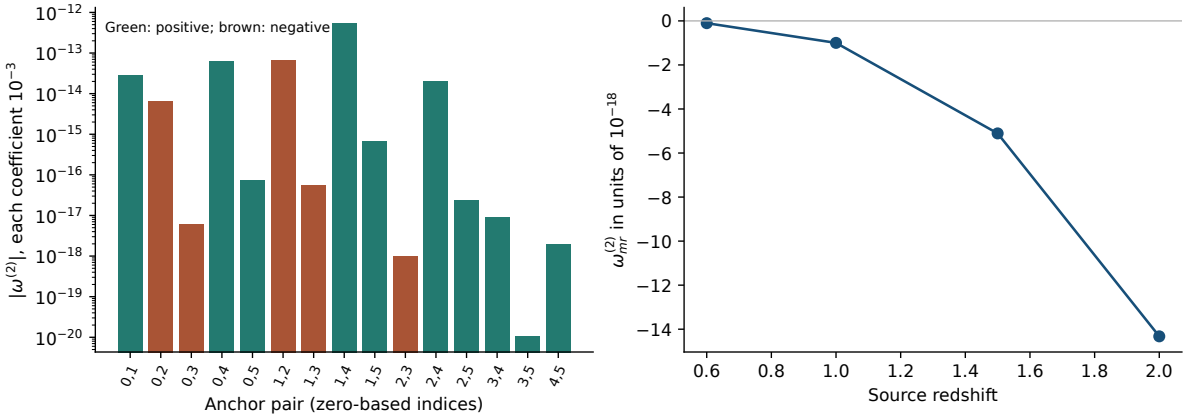


Figure 2: Conditional state responses on the round R1 branch. Left: all fifteen absolute pair responses at $z_s = 1$ for 10^{-3} in each indicated anchor coordinate; signs and the full matrix are in the supplied JSON. Coordinate magnitudes are not comparable physical energies. Right: smooth ordered rotation for the stated density pair across source redshifts; the marker at $z_s = 1$ matches Eq. (27). These are parallel-screen transfer coefficients, not detection forecasts.

The tiny density-pair result is retained. The larger response is a different coordinate probe with a different physical preparation; no amplitude is selected to match an observation.

For the density pair, 801-point ordered quadrature and the response ODE agree in $u^T \mathsf{B} v$ to 5.2×10^{-29}; refining the temporal provider from 801 to 1601 samples and tightening the propagator changes it by 1.3×10^{-28}. Direct finite matrix-ODE checks agree with the coefficient within 4.3×10^{-26} over the stated amplitude scales. The common-profile finite check is consistent with zero at 3.4×10^{-21} absolute map antisymmetry. Independent synthetic time-varying tides check the sign, amplitude convergence and zero-twist Wronskian without using the R1 response implementation.

For the illustrative density pair, sampled $\nu = 6.56 \times 10^{-6}$ gives a higher-even-order majorant about 5.11×10^{-27} in the specified first-order-tide ODE. Floating-point cancellation dominates the smallest direct differences; the larger reported differences are kept as numerical limitations. Arbitrary symmetric second-order test tides leave the limiting quadratic rotation unchanged, as the derivation requires. They are diagnostic forcings, not physical second-order fields.

On 161 time samples from $z_i = 2.1$ to zero, the reduced kinetic block has minimum eigenvalue 2.485×10^{-3} and the auxiliary block has maximum condition number 1.03×10^5. Original constraint residuals are at most 2.92×10^{-11} in the recorded scaled test, and the physical anchor reconstruction error is 2.23×10^{-16}. Sampled global angular bounds on $|\Phi|, |\Psi|$ across that full time interval are below 4.35×10^{-4} for the density pair and 1.90×10^{-3} for the momentum illustration. These support a weak perturbative example in the tested domain; they do not certify continuum stability, nonlinear admission or a unique physical state. A successful solve of an auxiliary block is not a ghost/stability proof.

As a fixed-lens illustration take reduced local convergence 0.45, shear amplitude 0.35 and axis 0.31 radians at $(z_l, z_s) = (0.3, 1)$. For the density pair Eq. (25) gives angular-source antisymmetric coefficient 6.82191×10^{-8}; centered finite differencing agrees to 1.93×10^{-15}. Its Fermat-weighted antisymmetric derivative vanishes to rounding, as required by Eq. (26). This is a representation check with prescribed local inputs, not an inferred astronomical rotation.

8 Scientific benefit and remaining physical freedom

The sharp bound supplies a genuine restriction that was absent from the supplied angular seed: shear with a structurally stationary nodal degree-two ray necessarily involves Hopf-charged modes. On a Hopf-fibre ray it is entirely charged; elsewhere its mixed sectors have a specified fraction and cannot preserve the same node under the auxiliary split spectral dynamics. This tells a subsequent nonround action calculation which state restrictions it must respect, rather than treating the round node as a generic Berger configuration.

The optical calculation supplies a state family instead of inventing a vacuum or aligning environmental data to obtain a desired number. Independent histories open a noncommuting transfer channel, but the established equivalent-lens freedom limits what that channel can mean without calibrated sources and lenses. The small coefficients in the examples give no basis for a detection forecast. No stochastic ensemble, survey likelihood or fresh observational validation is supplied.

The microscopic-to-R1 identification remains a specific missing object: an evaluated common-domain normalized action, background, constrained source pairing and metric/phase projection that identifies the exported BHSM field with the cosmological state and its admissible boundary data. The recovered Schur contracts and microscopic interface derivatives do not by themselves provide it. A complete squashed-spacetime application would additionally require a consistent nonround background, constraints, state evolution, parallel-screen ray and metric response. Substituting Berger eigenvalues into the isotropic R1 equations is not that derivation.

Relative to the known spectrum [3], this work proves the constrained fraction and its persistence consequence. Relative to the general lensing and second-order rotation treatments [7, 9], it evaluates a normalized conditional state operator and verifies its source-remapping limit. These are bounded theoretical/computational additions, not a new general lensing principle or the completion of BHSM. Their scientific value is that the assumptions and remaining freedoms are explicit enough to test.

A Frozen effective action and six-state temporal provider

This appendix specifies the conditional R1 realization used for the temporal response coefficients. It retains the coupled scalar, dust, and radiation system of the environmental and transport source calculations [1, 2]. It does not infer the microscopic BHSM state or its absolute normalization. The vendored source files and their hashes identify the precise implementation; only the background entries of the inherited parameter manifest are used.

A.1 Action, units, and background

The metric signature is $(-,+,+,+)$ and dots denote cosmic-time derivatives. In the code units $H_0 = M_{\rm Pl} = c = 1$, the effective action, up to its constant dimensional prefactor, is

$$S = \int d^4x\sqrt{-g}\left[\frac{\mathcal{R}}{2} + X - V(T) - \frac{2X}{\lambda}\Box T\right] + S_m + S_r, \qquad X = -\frac{1}{2}\nabla_\mu T\nabla^\mu T, \quad V(T) = V_0 - q_T T. \tag{28}$$

Thus the cubic term is $+(\nabla T)^2\Box T/\lambda$, with no additional factor of $1/2$. In the convention with dimensional cubic scale Λ, $G_3 = 2X/\Lambda^3$ and $\lambda = \Lambda^3/(M_{\rm Pl}H_0^2)$ after rescaling $T_{\rm phys} = M_{\rm Pl}T$ and time by H_0^{-1}. The fluids are minimally coupled, isentropic, irrotational perfect fluids with $w_m = 0$ and $w_r = 1/3$; the dust sound speed is exactly zero.

The frozen parameters are

$$\begin{aligned}(\Omega_{m0},\Omega_{r0},\Omega_{k0},\lambda,q_T) &= (0.31, 9\times10^{-5}, -0.018, 1, 3),\\ V_0 &= 2.3986073449708067.\end{aligned} \tag{29}$$

Here $K=-\Omega_{k0}=0.018$ is the spatial curvature in H_0^2/c^2 units; V_0 is in $M_{\rm Pl}^2H_0^2$ units. With $v=\dot T$, the background implementation uses

$$\begin{aligned} J &= v+6Hv^2/\lambda, & \dot J+3HJ &= q_T,\\ \rho_T &= \tfrac12 v^2+V+6Hv^3/\lambda, & 3(H^2+K/a^2) &= \rho_m+\rho_r+\rho_T, \end{aligned} \tag{30}$$

where $\rho_m=3\Omega_{m0}a^{-3}$ and $\rho_r=3\Omega_{r0}a^{-4}$. At $z=100$, the inherited background prescription sets $T=0$ and $J=2q_T/(9H)$, selecting the positive-v solution jointly with the Friedmann constraint. The stored V_0 comes from the original $H(a=1)=1$ normalization; this calculation neither repeats that shooting nor retunes it. Replaying the frozen input gives $H(a=1)=0.9999999999811999$, a normalization residual of -1.88×10^{-11}. The background is integrated in $N=\ln a$ with DOP853, relative tolerance 2×10^{-13}, absolute tolerance 10^{-15}, and maximum step 0.005.

A.2 Fluid source pairing and finite harmonic reduction

For each isentropic irrotational fluid the density-current convention is the Schutz–Sorkin form [14],

$$S_I=-\int d^4x\,[\sqrt{-g}\rho_I(n_I)+J_I^\mu\partial_\mu\ell_I],\quad n_I=\frac{\sqrt{-g_{\mu\nu}J_I^\mu J_I^\nu}}{\sqrt{-g}},\quad \rho_I\propto n_I^{1+w_I}. \tag{31}$$

Here J^μ is a vector density, $\mu_I=\partial\rho_I/\partial n_I$, $\dot{\bar\ell}_I=-\mu_I$, $\delta\ell_I=\mu_I v_I$, and $\delta J_I^0/\bar J_I^0=D_I$ at first order. Then $\delta n_I/\bar n_I=D_I-3\zeta$. Variation of the spatial current gives, to first order,

$$\frac{J_I^i}{\bar J_I^0}=a^{-2}\nabla_\gamma^i(v_I-\chi_{\rm sh}). \tag{32}$$

For clarity, before eliminating it put $s^i=J^i/\bar J^0$. The quadratic spatial-current terms divided by a^3 are $h_I[\frac12 h_{ij}(s^i+N^i)(s^j+N^j)-s^i\partial_i v_I]$. Their variation gives $h_{ij}(s^j+N^j)=\partial_i v_I$; substitution yields $h_I[-(\nabla v_I)^2/(2a^2)+\nabla\chi_{\rm sh}\cdot\nabla v_I/a^2]$. Harmonic integration gives the $-b_2v_I^2/2+b_2\chi_{\rm sh}v_I$ terms below. The temporal source pairing $-\delta J^0\partial_t(\mu_I v_I)$, integrated by parts, gives $a^3h_Iv_I\dot D_I$ because $\bar J^0$ is conserved. In the resulting density coordinate $\partial_t\ln(a^3h_I)=-3w_IH$.

Background tadpoles cannot be dropped independently. The fluid expansion contains $9p_I\zeta^2/2+3p_I\alpha\zeta-h_I\alpha D_I$. The gravity–scalar expansion on the total matter background contributes $-9p_{\rm tot}\zeta^2/2+3\rho_{\rm tot}\alpha\zeta$ relative to its vacuum-equation-simplified form. The sum is $-\sum_I h_I\alpha(D_I-3\zeta)$. The unitary-gauge cubic ADM density used for this bookkeeping is

$$N\sqrt{h}\left[\frac{{}^{(3)}R+K_{ij}K^{ij}-K_{\rm ext}^2}{2}+\frac{v^2}{2N^2}-V+\frac{2v^3K_{\rm ext}}{3\lambda N^3}\right], \tag{33}$$

with homogeneous lapse density $-3H^2/N+v^2/(2N)-NV+2Hv^3/(\lambda N^3)+3\kappa N$. The lapse and volume terms use the total Friedmann and pressure equations. The gravity–scalar curved-harmonic quadratic action is the specialization of the nonflat Horndeski reduction [12]; no flat-space replacement is made. Perturbations vanish at temporal variation endpoints; the closed spatial sphere has no boundary. These equations specify the reduction convention; the supplied residual tests independently check the finite polynomial and current elimination, not a second symbolic expansion of every covariant gravitational term. The recovered historical derivation and its original failed-comparison flags are preserved in the provenance supplement rather than silently erased.

For one real harmonic, $\int Q^2 dV_\gamma = 1$ and $\nabla_\gamma^2 Q = -8KQ$. The scalar-field uniform-T ADM convention uses lapse perturbation α, covariant shift $N_i = \nabla_i(\chi_{\rm sh} Q)$, and spatial curvature perturbation ζ. All variables below are harmonic coefficients. Define

$$\begin{gathered}\boldsymbol{x} = (\zeta, D_m, D_r)^T, \qquad D_I = \frac{\delta\rho_I}{\rho_I + p_I} + 3\zeta, \qquad \boldsymbol{u} = (\alpha, \chi_{\rm sh}, v_m, v_r)^T, \\ \kappa = K/a^2, \qquad b_2 = 8\kappa, \qquad s_2 = 5\kappa.\end{gathered} \tag{34}$$

The symbols v_m, v_r here denote ADM fluid velocity potentials, not the background field velocity v. Put $h_I = \rho_I + p_I$ and

$$\Theta = H - v^3/\lambda, \qquad \Sigma = \tfrac{1}{2}v^2 + 12Hv^3/\lambda - 3H^2. \tag{35}$$

After algebraically eliminating the spatial fluid-current auxiliaries and performing the declared integration by parts, the implemented polynomial is

$$\begin{aligned}\frac{L_2}{a^3} =& -3\dot\zeta^2 + s_2\zeta^2 + \Sigma\alpha^2 + 2b_2\Theta\alpha\chi_{\rm sh} - 2b_2\dot\zeta\chi_{\rm sh} \\ &+ 6\Theta\alpha\dot\zeta + 2s_2\alpha\zeta - b_2\kappa\chi_{\rm sh}^2 + \sum_{I=m,r} h_I\ell_I, \\ \ell_I =& \, v_I\dot D_I - \frac{b_2}{2}v_I^2 + b_2\chi_{\rm sh}v_I - \frac{w_I}{2}(D_I - 3\zeta)^2 - \alpha(D_I - 3\zeta).\end{aligned} \tag{36}$$

This equation specifies every finite Hessian entry used by the provider. Write it as

$$L_2/a^3 = \tfrac{1}{2}\boldsymbol{u}^T A\boldsymbol{u} + \boldsymbol{u}^T(B\dot{\boldsymbol{x}} + D\boldsymbol{x}) + \tfrac{1}{2}\dot{\boldsymbol{x}}^T M_0\dot{\boldsymbol{x}} + \tfrac{1}{2}\boldsymbol{x}^T E\boldsymbol{x}. \tag{37}$$

The auxiliary equations and their Schur reduction give

$$\begin{aligned}&\boldsymbol{u} = -A^{-1}(B\dot{\boldsymbol{x}} + D\boldsymbol{x}), \\ &M = M_0 - B^T A^{-1}B, \qquad L = -B^T A^{-1}D, \qquad V_2 = E - D^T A^{-1}D.\end{aligned} \tag{38}$$

The reduced canonical momentum is $\boldsymbol{p} = a^3(M\dot{\boldsymbol{x}} + L\boldsymbol{x})$. Consequently the six-dimensional state $\boldsymbol{w} = (\boldsymbol{x}, \boldsymbol{p})^T$ evolves by

$$\dot{\boldsymbol{x}} = M^{-1}(a^{-3}\boldsymbol{p} - L\boldsymbol{x}), \qquad \dot{\boldsymbol{p}} = a^3(L^T\dot{\boldsymbol{x}} + V_2\boldsymbol{x}). \tag{39}$$

Dividing these equations by H defines $d\boldsymbol{w}/dN = F(N)\boldsymbol{w}$ and the full fundamental matrix $U' = FU$, $U(N_i) = I_6$. No projection or restart onto an isolated two-state sector is performed. The default propagator tolerances are $(2\times10^{-10}, 2\times10^{-12})$, with maximum N step 0.01; the refinement uses $(2\times10^{-12}, 2\times10^{-14})$ and step 0.005.

A.3 Physical anchor and optical potential rows

At $z_i = 2.1$, $N_i = -\ln 3.1$, the six specified coordinates are

$$\begin{gathered}\boldsymbol{y} = (\zeta, \Pi_{2,\rm ref}, \delta_m^{\rm ADM}, \delta_r^{\rm ADM}, v_m^{\rm ADM}, v_r^{\rm ADM})_i^T, \\ \Pi_{2,\rm ref} = 2a_i^3 G_{S,2}^{\rm ref}\dot\zeta_i, \qquad G_{S,2}^{\rm ref} = \frac{5(r+3)}{r+8}, \qquad r = \frac{\Sigma}{\Theta^2}.\end{gathered} \tag{40}$$

The reference factor is the inherited gravity–scalar normalization; $\Pi_{2,\rm ref}$ is not identified with the coupled canonical momentum p_ζ. The current action rows implement $\delta_m^{\rm ADM} = D_m - 3\zeta$, $\delta_r^{\rm ADM} = \frac{4}{3}(D_r - 3\zeta)$, and the reconstructed velocity potentials. If $\mathcal{B}_i\boldsymbol{w}_i = \boldsymbol{y}$ denotes these six rows, the initial canonical state is obtained by solving this full system, $\boldsymbol{w}_i = \mathcal{B}_i^{-1}\boldsymbol{y}$. All six coordinates are retained independently for every spatial harmonic.

The Newtonian metric convention is $ds^2 = -(1+2\Phi)dt^2 + a^2(1-2\Psi)\gamma_{ij}dx^i dx^j$. For one harmonic, $\Phi = \phi Q$ and $\Psi = \psi Q$, with reconstructed rows

$$\begin{aligned} \phi &= \alpha + \dot{\chi}_{\rm sh}, \qquad \psi = -\zeta - H\chi_{\rm sh}, \\ \delta_I^{\rm N} &= (1+w_I)(D_I - 3\zeta - 3H\chi_{\rm sh}), \qquad v_I^{\rm N} = v_I^{\rm ADM} - \chi_{\rm sh}. \end{aligned} \tag{41}$$

The derivative of $\chi_{\rm sh}$ includes both the background dependence of its reconstruction row and the propagated state. The code evaluates the former by a complex-step directional derivative. Multiplication of these metric rows by $U(N, N_i)\mathcal{B}_i^{-1}$ gives the six columns $\phi_j(N), \psi_j(N)$ used by the optical integrals. Cubic interpolation uses 801 temporal samples, repeated with 1601 samples in the refinement. The reported reconstruction, action-constraint, and independent optical checks establish numerical consistency of this declared finite system. They do not establish a general stability theorem, select its initial state, or supply the missing microscopic BHSM preparation.

B Reproduction, provenance and disclosure

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Author contribution and assistance. Norman P. Carberry provided the research direction and remains responsible for scientific review and approval. AI-assisted tools developed code and exposition and performed separate internal proof and claim reviews. These reviews are reproducibility checks, not external human peer review.

Funding and competing interests. Final declarations require author confirmation before any submission. Independent status does not determine either declaration.

Data availability. All examples are synthetic forward calculations. Source, inputs, proofs, output matrices and replay commands are supplied; no astronomical detection is asserted.

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